

K=2 Lazy Ring with One Directed Shortcut: Beta Sweep for the Double Peak

Conclusion

For the tested case $N = 100$, $q = 2/3$, $u = 6$, $v = 56$, and $n_0 = 1, \dots, 6$, the proposed boundary is

$$\beta_c = \frac{2q}{(1-q)N} = 0.04.$$

I use R3 as the main criterion for a visually resolved double peak. Let $t_1 < t_2$ be the first two filtered local-maximum times, with $h_1 = f(t_1)$ and $h_2 = f(t_2)$, and let $h_v = f(t_v)$ be the minimum value of $f(t)$ between those two peaks. Define

$$\rho = \frac{h_v}{\min(h_1, h_2)}.$$

The criteria used below are:

- R1 : two filtered local maxima, separated in time and with balanced heights,
- R2 : $R1 + h_v / \max(h_1, h_2) \leq 0.9$,
- R3 : $R1 + \rho = h_v / \min(h_1, h_2) \leq 0.8$,
- R4 : $R1 + \rho = h_v / \min(h_1, h_2) \leq 0.6$.

R2 is a permissive “there is a dip” diagnostic; R3 is the main visual double-peak criterion because it requires the valley to be clearly below the smaller peak.

The assumption behind the proposed boundary is that the transition can be captured by an n_0 -independent necessary condition in β . The sweep supports this assumption only in the visual sense: under R3, no clear double peak remains at or above β_c . The reference case $n_0 = 1$, $\beta = 0.01$ does give a clear double peak, but at $\beta = 0.04$ the observed values are already $\rho \simeq 0.887\text{--}0.891$ for $n_0 = 1, 2, 3$, so the feature is better described as a shallow shoulder or marginal second peak.

The proposed boundary is therefore supported as a cutoff for a visually clear double peak, but not as a strict cutoff for the existence of two mathematical local maxima. Under the weaker R1/R2 rules, two local maxima can still be detected above β_c for $n_0 = 1, 2, 3$, up to about $\beta = 0.060\text{--}0.062$. The starting point n_0 matters, so the expression cannot be the whole condition.

Question

I test the proposed condition for the first-passage distribution on a lazy nearest-neighbour ring with one directed shortcut. The model is

$$N = 100, \quad K = 2, \quad q = \frac{2}{3}, \quad u = 6, \quad v = 56,$$

with paper-indexed sites. At the shortcut source, a fraction of the lazy self-loop probability is moved to the directed shortcut:

$$P(u \rightarrow v) = \beta(1 - q), \quad P(u \rightarrow u) = (1 - \beta)(1 - q).$$

The sweep uses start positions $n_0 = 1, \dots, 6$ and beta values from 0 to 0.08, with extra points at the reference values. The default target is the named endpoint $v = 56$; the script can be rerun with another target convention using `--target=paper`.

The proposed boundary is

$$\beta_c = \frac{2q}{(1-q)N} = 0.04.$$

Framework

For each (n_0, β) I compute $f(t)$ up to $t_{\max} = 12000$. Because the phrase “double peak” can mean either “two mathematical local maxima” or “two peaks clearly visible by eye”, I record several definitions. The visual definitions

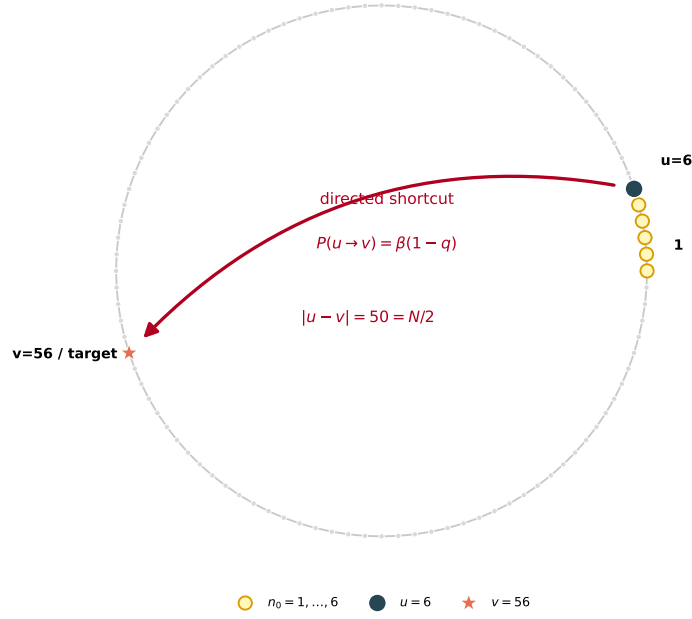


Figure 1: Concrete geometry of the tested ring. The grey circle is the $N = 100$, $K = 2$ nearest-neighbour ring; yellow points show the swept starts $n_0 = 1, \dots, 6$; the red curved arrow is the directed shortcut $u = 6 \rightarrow v = 56$, whose endpoint is also the absorbing target.

use $h_v / \min(h_1, h_2)$ because a curve only looks like two peaks if the valley is below both peaks, especially the smaller one.

The quantities used in the rules are:

$$t_1 < t_2, \quad h_1 = f(t_1), \quad h_2 = f(t_2), \quad t_v = \arg \min_{t_1 < t < t_2} f(t), \quad h_v = f(t_v).$$

Here t_1 and t_2 are the first two filtered local-maximum times, t_v is the intervening valley time, h_2/h_1 measures the relative peak heights, $h_v / \max(h_1, h_2)$ measures whether there is any valley below the larger peak, and $h_v / \min(h_1, h_2)$ measures whether the valley is visibly below both peaks.

The operational rules are:

- R1 Two local maxima:** two filtered local peaks with $t_2 - t_1 \geq 10$ and balanced peak heights $0.1 \leq h_2/h_1 \leq 10$.
- R2 Permissive valley:** R1 plus $h_v / \max(h_1, h_2) \leq 0.9$.
- R3 Visual double peak:** R1 plus $\rho = h_v / \min(h_1, h_2) \leq 0.8$.
- R4 Strong visual double peak:** R1 plus $\rho = h_v / \min(h_1, h_2) \leq 0.6$.

The difference between R2 and R3 is intentional. R2 uses $h_v / \max(h_1, h_2)$, so it only asks whether the curve has a dip below the larger of the two peaks. This is a permissive diagnostic: it confirms that a valley exists, but it can still count cases where the valley is almost as high as the smaller peak. Such cases may have two mathematical local maxima without looking like two clearly separated peaks.

R3 uses $h_v / \min(h_1, h_2)$ instead. This is the more relevant visual test because a double peak is only clearly resolved if the valley is below both peaks, especially below the smaller one. The cutoff $\rho \leq 0.8$ is therefore used as the main “visible double peak” rule: it requires a valley at least about 20% below the smaller peak. R4 keeps the same logic but uses the stricter cutoff $\rho \leq 0.6$ to identify only strongly separated cases.

This threshold should be read as a practical visual convention rather than an absolute mathematical constant. If the visual rule were relaxed toward $\rho \leq 0.9$, the boundary case $\beta = \beta_c$ would become borderline, because the measured values there are $\rho \simeq 0.887\text{--}0.891$ for $n_0 = 1, 2, 3$. If the rule is tightened to R4, the detected double-peak region shrinks to smaller beta values. This sensitivity is why R1/R2, R3, and R4 are reported separately.

Main Readout

For the reference point mentioned in the request, $n_0 = 1$ and $\beta = 0.01$, the sweep gives a clear double peak with

$$t_1 = 37, \quad t_v = 226, \quad t_2 = 1077,$$

Table 1: Summary of the fixed-shortcut sweep. “Double at/above β_c ” counts grid points satisfying the R3 visual double-peak rule and $\beta \geq 0.04$.

n_0	target	β_c	any double	max β double	double at/above β_c
1	56	0.040	yes	0.030	0
2	56	0.040	yes	0.030	0
3	56	0.040	yes	0.030	0
4	56	0.040	no	—	0
5	56	0.040	no	—	0
6	56	0.040	no	—	0

and peak-height ratio $h_2/h_1 \approx 1.73$.

Observed Phenomenology

Table 2: Observed behaviour under different double-peak definitions. Here ρ_c is $\rho = h_v / \min(h_1, h_2)$ at $\beta_c = 0.04$ for $n_0 = 1, 2, 3$. The range 0.887–0.891 means that the valley is already close to the smaller peak at the proposed boundary.

Definition	Cases with $\beta \geq \beta_c$	Maximum passing β	ρ_c
two local maxima	34	n0=1: 0.062; n0=2: 0.060; n0=3: 0.060; n0=4: none; n0=5: none; n0=6: none	0.887-0.891
permissive valley	34	n0=1: 0.062; n0=2: 0.060; n0=3: 0.060; n0=4: none; n0=5: none; n0=6: none	0.887-0.891
visual double peak	0	n0=1: 0.030; n0=2: 0.030; n0=3: 0.030	0.887-0.891
strong visual double peak	0	n0=1: 0.018; n0=2: 0.018; n0=3: 0.018	0.887-0.891

The different rules separate two effects. Under local-maximum rules, two maxima can still be detected above β_c for $n_0 = 1, 2, 3$. Under the visual rules based on $h_v / \min(h_1, h_2)$, no clear double peak appears at or above β_c . At $\beta = 0.04$, the detector can still find two local maxima for $n_0 = 1, 2, 3$, but $h_v / \min(h_1, h_2) \approx 0.89$, so the valley is shallow and the profile is visually closer to a shoulder or marginal second peak than to a clearly separated double peak.

Visual Evidence

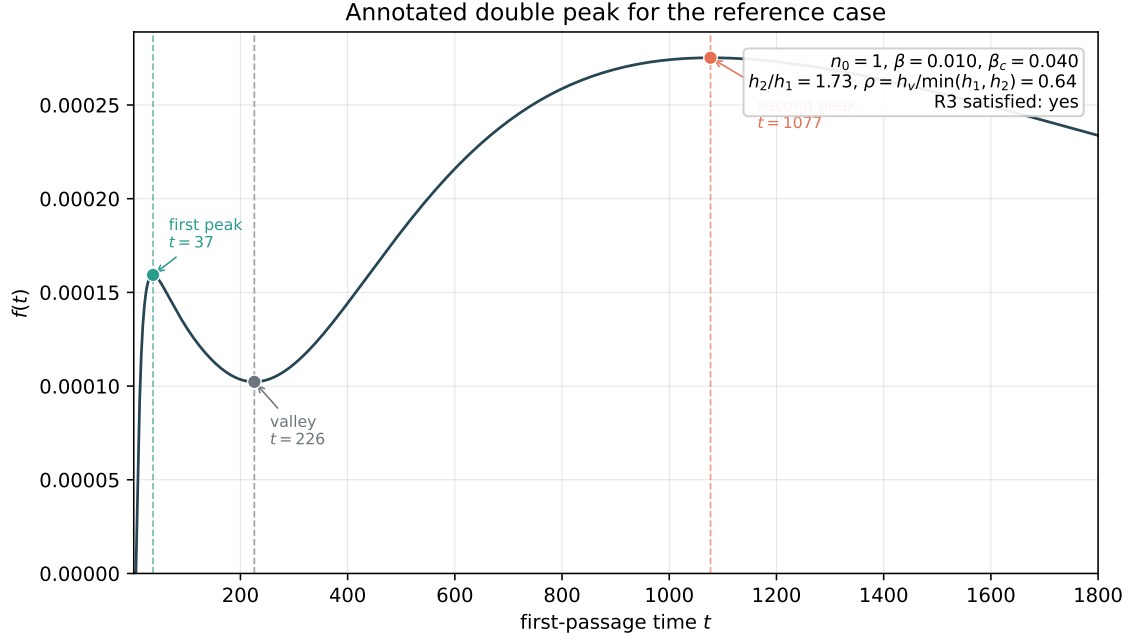


Figure 2: The concrete double-peak curve for the reference case $n_0 = 1$, $\beta = 0.01$. The first peak, valley, and second peak are marked explicitly.

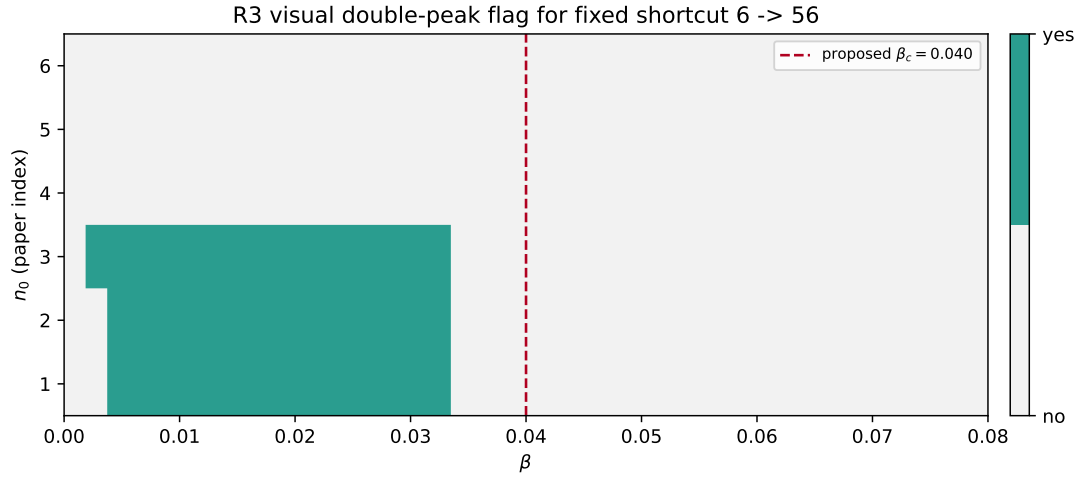


Figure 3: R3 visual double-peak flag over the (n_0, β) grid. The dashed red line is the proposed $\beta_c = 0.04$.

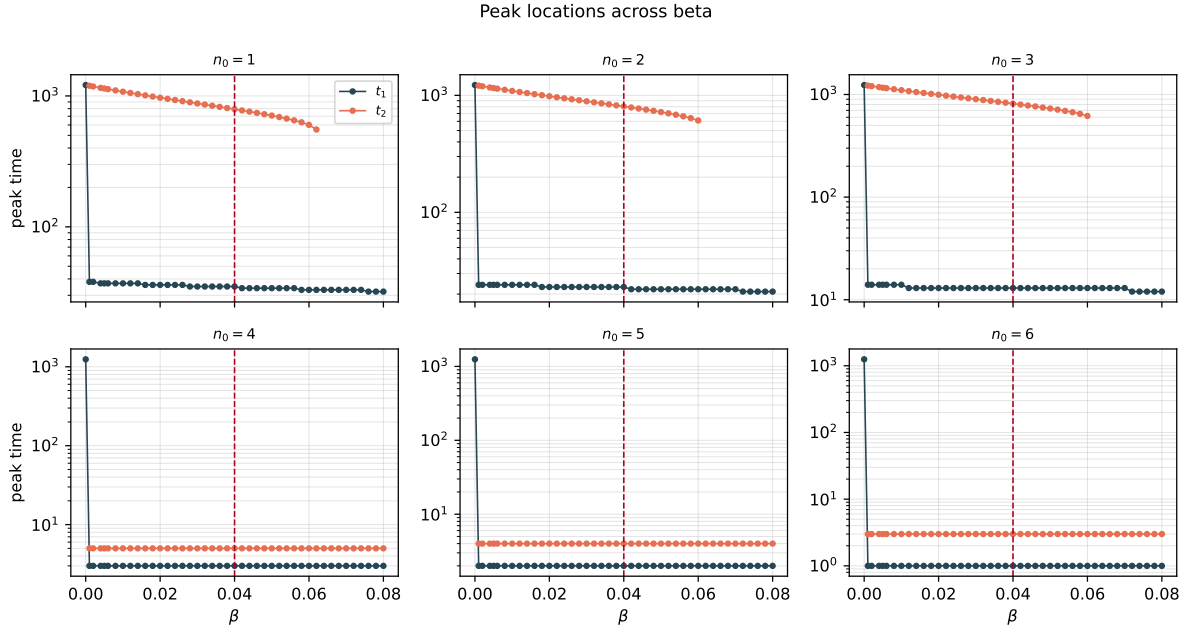


Figure 4: Detected peak locations across beta. The late peak remains separated from the early peak for $n_0 = 1, 2, 3$ beyond β_c on this grid.

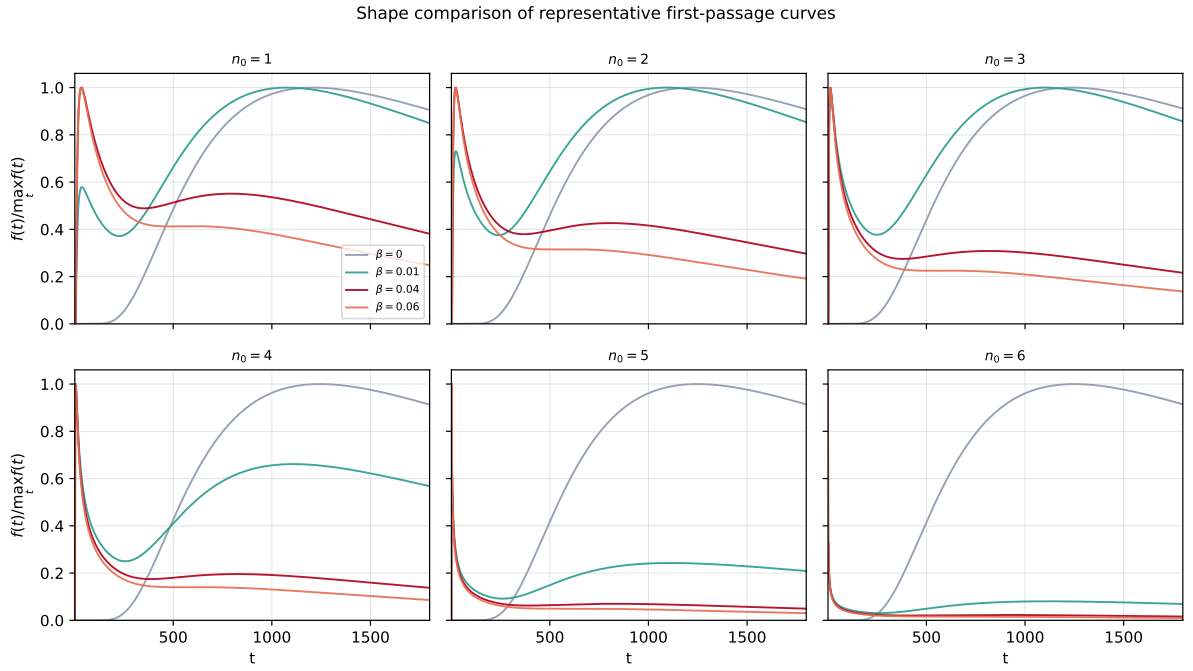


Figure 5: Representative first-passage curve shapes for selected beta values. Each curve is normalized by its own maximum to make the peak and valley geometry visible on a linear scale.

Interpretation

Under the literal target convention $target = v = 56$, the sweep does not support the strong statement that $\beta > \beta_c$ rules out *any two local maxima*. In the default grid, local maxima can persist up to $\beta \approx 0.062$ for $n_0 = 1$ and up to $\beta \approx 0.060$ for $n_0 = 2, 3$. However, under the more appropriate visual criterion $h_v / \min(h_1, h_2) \leq 0.8$, no clear double peak survives at or above $\beta_c = 0.04$.

This agrees with the caveat that the expression cannot be the whole story: the starting point n_0 materially changes the outcome, so any complete condition must include the geometry of the start, shortcut source, shortcut destination, and target.

Reproducibility

Run from the repository root:

```
.venv/bin/python research/reports/ring_lazy_jump_ext_rev2/code/\
luca_k2_fixed_shortcut_sweep.py
```

The run writes the config, metrics, summary table, selected curves, and all figures under this report's **artifacts/** directory.